

# Vector Calculus

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(adapted from Fieller (2016), [Section 7.2](#))

This section covers derivatives of functions of vectors and matrices. Linear algebra prerequisites — including vectors, matrices, transpose, dot product, and quadratic forms — are covered in [Linear Algebra](#).

Let  $\tilde{x}$  and  $\tilde{\beta}$  be column vectors of length  $p$  (see [column vector](#) and [dot product](#)).

**Definition 0.1** (Vector derivative). If  $f(\tilde{\beta})$  is a scalar-valued function of a  $p \times 1$  vector  $\tilde{\beta}$ , such as  $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$ , then its **vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) \\ \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) \\ \vdots \\ \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

## Video lecture

Hutchinson's [Gradients Refresher](#) (17 min) covers gradients and partial derivatives, the ideas behind this section (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

## Checking a gradient by its shape

The gradient always has the same shape as the input it is taken with respect to, because it holds exactly one partial derivative per component of that input:

Table 1: The shape of a gradient follows the shape of what it differentiates with respect to.

| $f$ is a function of                  | its gradient is                           |
|---------------------------------------|-------------------------------------------|
| a single number                       | a single number — the ordinary derivative |
| a vector in $\mathbb{R}^p$            | a vector in $\mathbb{R}^p$                |
| a matrix in $\mathbb{R}^{m \times n}$ | a matrix in $\mathbb{R}^{m \times n}$     |

The table is the cheapest check there is on a gradient just worked out by hand: if the shapes disagree, something went wrong. It is the shape rule of matrix multiplication again, applied to calculus.

The course website's version of this page has an interactive plot [here](#).

**Exercise 0.1** (Take a gradient by hand). Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be

$$f(\tilde{w}) = 3w_1w_2 + w_2^3 - 5w_1$$

1. Compute  $\partial f / \partial w_1$ , treating  $w_2$  as a constant.
2. Compute  $\partial f / \partial w_2$ , treating  $w_1$  as a constant.
3. Assemble  $\nabla_{\tilde{w}} f$  and evaluate it at  $\tilde{w} = [2 \ -1]^\top$ .
4. Check the shape of your answer against Table 1.

#### Solution

*Solution 0.1. 1.* Differentiate with respect to  $w_1$ , holding  $w_2$  fixed. The first term is the constant  $3w_2$  times  $w_1$ , so it contributes  $3w_2$ . The second term has no  $w_1$  in it, so it contributes 0. The third term is  $-5$  times  $w_1$ , so it contributes  $-5$ :

$$\frac{\partial f}{\partial w_1} = 3w_2 + 0 - 5 = 3w_2 - 5$$

**2.** Now with respect to  $w_2$ , holding  $w_1$  fixed. The first term is the constant  $3w_1$  times  $w_2$ , contributing  $3w_1$ . The second term contributes  $3w_2^2$ . The third has no  $w_2$  in it, contributing 0:

$$\frac{\partial f}{\partial w_2} = 3w_1 + 3w_2^2 + 0 = 3w_1 + 3w_2^2$$

**3.** Stack the two, in the order the coordinates are numbered:

$$\nabla_{\tilde{w}} f(\tilde{w}) = \begin{bmatrix} 3w_2 - 5 \\ 3w_1 + 3w_2^2 \end{bmatrix}$$

The gradient is a vector-valued *function* of  $\tilde{w}$ , not a single vector. At  $\tilde{w} = [2 \ -1]^\top$ ,

$$\nabla_{\tilde{w}} f = \begin{bmatrix} 3(-1) - 5 \\ 3(2) + 3(-1)^2 \end{bmatrix} = \begin{bmatrix} -3 - 5 \\ 6 + 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 9 \end{bmatrix}$$

4. The input was a vector in  $\mathbb{R}^2$  and so is the answer, as Table 1 requires. Note the signs: from this point, increasing  $w_1$  *decreases*  $f$  while increasing  $w_2$  increases it, so the uphill direction is neither axis.

**Definition 0.2** (Row-vector derivative). If  $f(\tilde{\beta})$  is a scalar-valued function of a  $p \times 1$  vector  $\tilde{\beta}$ , such as  $f(\tilde{\beta}) = \tilde{x}^\top \tilde{\beta}$ , then its **row-vector derivative** is:

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \begin{bmatrix} \frac{\partial}{\partial \beta_1} f(\tilde{\beta}) & \frac{\partial}{\partial \beta_2} f(\tilde{\beta}) & \cdots & \frac{\partial}{\partial \beta_p} f(\tilde{\beta}) \end{bmatrix}$$

**Theorem 0.1** (Row and column derivatives are transposes).

$$\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = \left( \frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) \right)^\top$$

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \left( \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) \right)^\top$$

### **i** Proof

*Proof.* By Definition 0.1 and Definition 0.2, entry  $j$  of both  $\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta})$  and  $\frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta})$  is  $\frac{\partial}{\partial \beta_j} f(\tilde{\beta})$ ; the first is a  $p \times 1$  column and the second a  $1 \times p$  row with the same entries in the same order, so each is the transpose of the other.  $\square$

**Example 0.1** (Row and column derivatives of a linear function). For  $f(\tilde{\beta}) = 3\beta_1 + 5\beta_2$ :

$$\frac{\partial}{\partial \tilde{\beta}} f(\tilde{\beta}) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \frac{\partial}{\partial \tilde{\beta}^\top} f(\tilde{\beta}) = [3 \quad 5],$$

and each is the transpose of the other.

**Definition 0.3** (Derivative of a vector-valued function). If  $\tilde{y} = \tilde{y}(\tilde{\beta}) = (y_1, \dots, y_q)^\top$  is a  $q \times 1$  vector-valued function of the  $p \times 1$  vector  $\tilde{\beta}$ , its **derivative with respect to  $\tilde{\beta}$**  is the  $p \times q$  matrix whose  $(i, j)$  entry is

$$\left[ \frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top \right]_{ij} \stackrel{\text{def}}{=} \frac{\partial}{\partial \beta_i} y_j, \quad i = 1, \dots, p, \quad j = 1, \dots, q.$$

These notes use this *denominator layout* throughout: rows index the entries of  $\tilde{\beta}$  (the denominator) and columns index the entries of  $\tilde{y}$  (the numerator), so column  $j$  is the vector derivative  $\frac{\partial}{\partial \tilde{\beta}} y_j$  (Definition 0.1). Both  $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}$  and  $\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top$  denote this  $p \times q$  matrix.

Some sources use the transposed, *numerator layout*, in which the derivative is the  $q \times p$  Jacobian matrix with  $(j, i)$  entry  $\frac{\partial}{\partial \beta_i} y_j$ . Check a source's layout before combining its formulas with these.

**Example 0.2** (Differentiating a  $3 \times 1$  function of a  $2 \times 1$  vector). Let  $\tilde{\beta} = (\beta_1, \beta_2)^\top$  ( $p = 2$ ) and  $\tilde{y}(\tilde{\beta}) = (\beta_1^2, \beta_1\beta_2, 3\beta_2)^\top$  ( $q = 3$ ). Then

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{y}^\top}_{2 \times 3} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} \beta_1^2 & \frac{\partial}{\partial \beta_1} \beta_1\beta_2 & \frac{\partial}{\partial \beta_1} 3\beta_2 \\ \frac{\partial}{\partial \beta_2} \beta_1^2 & \frac{\partial}{\partial \beta_2} \beta_1\beta_2 & \frac{\partial}{\partial \beta_2} 3\beta_2 \end{bmatrix} = \begin{bmatrix} 2\beta_1 & \beta_2 & 0 \\ 0 & \beta_1 & 3 \end{bmatrix}$$

**Definition 0.4** (Constant). A  $q \times 1$  vector  $\tilde{x}$  is **constant with respect to** the  $p \times 1$  vector  $\tilde{\beta}$  if its derivative (Definition 0.3) is zero:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{p \times q} = \mathbf{0}_{p \times q}$$

**Example 0.3** (A constant vector). Let  $\tilde{\beta} = (\beta_1, \beta_2)^\top$  and  $\tilde{x} = (3, 5)^\top$ , so  $x_1 = 3$  and  $x_2 = 5$  do not depend on  $\tilde{\beta}$ . Expanding  $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top$  into its matrix of scalar partial derivatives (Definition 0.3) and evaluating each entry:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top}_{2 \times 2} = \frac{\partial}{\partial \tilde{\beta}} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} x_1 & \frac{\partial}{\partial \beta_1} x_2 \\ \frac{\partial}{\partial \beta_2} x_1 & \frac{\partial}{\partial \beta_2} x_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial \beta_1} 3 & \frac{\partial}{\partial \beta_1} 5 \\ \frac{\partial}{\partial \beta_2} 3 & \frac{\partial}{\partial \beta_2} 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{0}_{2 \times 2}$$

Every entry is the derivative of a constant, so  $\frac{\partial}{\partial \tilde{\beta}} \tilde{x}^\top = \mathbf{0}_{2 \times 2}$  and  $\tilde{x}$  is constant with respect to  $\tilde{\beta}$  (Definition 0.4).

**Theorem 0.2** (Derivative of a dot product). *If  $\tilde{x}$  is constant with respect to  $\tilde{\beta}$ , then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta})}_{p \times 1} = \underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta} \cdot \tilde{x})}_{p \times 1} = \underbrace{\tilde{x}}_{p \times 1}$$

### **i** Proof

*Proof.*

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) &= \begin{bmatrix} \frac{\partial}{\partial \beta_1}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \frac{\partial}{\partial \beta_2}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \\ \vdots \\ \frac{\partial}{\partial \beta_p}(x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p) \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \\ &= \tilde{x} \end{aligned}$$

□

**Example 0.4** (Derivative of a dot product). Let  $\tilde{x} = (3, 5)^\top$  (constant with respect to  $\tilde{\beta}$ ; see Example 0.3) and  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ . Then  $\tilde{x} \cdot \tilde{\beta} = 3\beta_1 + 5\beta_2$ , and by Theorem 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta})}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Verifying entry-wise:

$$\frac{\partial}{\partial \tilde{\beta}}(3\beta_1 + 5\beta_2) = \begin{pmatrix} \frac{\partial}{\partial \beta_1}(3\beta_1 + 5\beta_2) \\ \frac{\partial}{\partial \beta_2}(3\beta_1 + 5\beta_2) \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Both methods agree.

**Theorem 0.3** (Product rule for dot-products). *If  $\tilde{a} = \tilde{a}(\tilde{x})$  and  $\tilde{b} = \tilde{b}(\tilde{x})$  are differentiable  $p \times 1$  vector functions of  $\tilde{x}$ , then:*

$$\frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \cdot \underbrace{\tilde{a}}_{p \times 1} \cdot \underbrace{\tilde{b}}_{p \times 1} = \left( \frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{a}^\top}_{1 \times p} \right) \underbrace{\tilde{b}}_{p \times 1} + \left( \frac{\partial}{\partial \tilde{x}} \underbrace{\tilde{x}}_{p \times 1} \underbrace{\tilde{b}^\top}_{1 \times p} \right) \underbrace{\tilde{a}}_{p \times 1}$$

**i** Proof

*Proof.* Entry-wise, for  $i = 1, \dots, p$ :

$$\begin{aligned} \left[ \frac{\partial}{\partial \tilde{x}} (\tilde{a} \cdot \tilde{b}) \right]_i &= \frac{\partial}{\partial x_i} \sum_{k=1}^p a_k b_k \\ &= \sum_{k=1}^p \left( b_k \frac{\partial}{\partial x_i} a_k + a_k \frac{\partial}{\partial x_i} b_k \right) \\ &= \left[ \left( \frac{\partial}{\partial \tilde{x}} \tilde{a}^\top \right) \tilde{b} \right]_i + \left[ \left( \frac{\partial}{\partial \tilde{x}} \tilde{b}^\top \right) \tilde{a} \right]_i \end{aligned}$$

□

**Example 0.5** (Example of the dot-product rule). Apply Theorem 0.3 with the vector  $\tilde{\beta} = (\beta_1, \beta_2)^\top$  in the role of  $\tilde{x}$ . Let  $\tilde{a}(\tilde{\beta}) = (\beta_1, \beta_1 \beta_2)^\top$  and  $\tilde{b}(\tilde{\beta}) = (\beta_2, \beta_1)^\top$ . Then:

$$\tilde{a} \cdot \tilde{b} = \beta_1 \cdot \beta_2 + \beta_1 \beta_2 \cdot \beta_1 = \beta_1 \beta_2 + \beta_1^2 \beta_2$$

By direct calculation:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{a} \cdot \tilde{b})}_{2 \times 1} = \frac{\partial}{\partial \tilde{\beta}} (\beta_1 \beta_2 + \beta_1^2 \beta_2) = \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1 + \beta_1^2 \end{pmatrix}$$

By the product rule (Theorem 0.3), using  $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{a}^\top}_{2 \times 2} = \begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}$  and  $\underbrace{\frac{\partial}{\partial \tilde{\beta}} \tilde{b}^\top}_{2 \times 2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ :

$$\begin{aligned} \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{a} \cdot \tilde{b})}_{2 \times 1} &= \underbrace{\begin{pmatrix} 1 & \beta_2 \\ 0 & \beta_1 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_2 \\ \beta_1 \end{pmatrix}}_{2 \times 1} + \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{2 \times 2} \underbrace{\begin{pmatrix} \beta_1 \\ \beta_1 \beta_2 \end{pmatrix}}_{2 \times 1} \\ &= \begin{pmatrix} \beta_2 + \beta_1 \beta_2 \\ \beta_1^2 \end{pmatrix} + \begin{pmatrix} \beta_1 \beta_2 \\ \beta_1 \end{pmatrix} \\ &= \begin{pmatrix} \beta_2 + 2\beta_1 \beta_2 \\ \beta_1^2 + \beta_1 \end{pmatrix} \end{aligned}$$

Both methods agree.

**Theorem 0.4** (Derivative of a linear map). If  $\mathbf{A}$  is an  $m \times p$  matrix that is constant with respect to  $\tilde{\beta}$ , then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta})}_{p \times m} = \underbrace{\mathbf{A}^\top}_{p \times m}$$

### **i** Proof

*Proof.* For entry  $(i, j)$ , where row  $i$  indexes the denominator  $\tilde{\beta}$  (see Definition 0.3) and column  $j$  indexes the numerator  $\mathbf{A}\tilde{\beta}$ :

$$\begin{aligned} \left[ \frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta}) \right]_{ij} &= \frac{\partial}{\partial \beta_i}(\mathbf{A}\tilde{\beta})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^p a_{jk} \beta_k \\ &= a_{ji} \\ &= [\mathbf{A}^\top]_{ij} \end{aligned}$$

□

**Example 0.6** (Derivative of a linear map). Let  $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$  ( $1 \times 2$ ) and  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ . Then  $\mathbf{A}\tilde{\beta} = 2\beta_1 + 3\beta_2$ , and by Theorem 0.4:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta})}_{2 \times 1} = \underbrace{\mathbf{A}^\top}_{2 \times 1} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

**Theorem 0.5** (Vector-derivative of a matrix-vector product). *If  $\mathbf{A}$  is an  $m \times q$  matrix that is constant with respect to  $\tilde{\beta}$ , and  $\tilde{v} = \tilde{v}(\tilde{\beta})$  is a  $q \times 1$  vector that depends on the  $p \times 1$  vector  $\tilde{\beta}$ , then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{v})}_{p \times m} = \underbrace{\left( \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{A}^\top}_{q \times m}$$

*This result generalizes Theorem 0.4, which is the special case  $\tilde{v} = \tilde{\beta}$  (so that  $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$  and  $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{A}\tilde{\beta}) = \mathbf{A}^\top$ ).*

**i** Proof

*Proof.* For entry  $(i, j)$ , where row  $i$  indexes the denominator  $\tilde{\beta}$  and column  $j$  indexes the numerator  $\mathbf{A}\tilde{v}$  (see Definition 0.3):

$$\begin{aligned} \left[ \frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{v}) \right]_{ij} &= \frac{\partial}{\partial \beta_i} (\mathbf{A}\tilde{v})_j \\ &= \frac{\partial}{\partial \beta_i} \sum_{k=1}^q a_{jk} v_k \\ &= \sum_{k=1}^q a_{jk} \frac{\partial}{\partial \beta_i} v_k \\ &= \sum_{k=1}^q \left[ \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right]_{ik} [\mathbf{A}^\top]_{kj} \\ &= \left[ \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{A}^\top \right]_{ij} \end{aligned}$$

□

**Example 0.7** (Vector-derivative of a matrix-vector product). Let  $\mathbf{A} = \begin{pmatrix} 2 & 3 \end{pmatrix}$  ( $1 \times 2$ , constant) and  $\tilde{v}(\tilde{\beta}) = (\beta_1^2, \beta_2^2)^\top$ . Then  $\mathbf{A}\tilde{v} = 2\beta_1^2 + 3\beta_2^2$ . By Theorem 0.5:

$$\begin{aligned} \underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\tilde{v})}_{2 \times 1} &= \begin{pmatrix} 2\beta_1 & 0 \\ 0 & 2\beta_2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 4\beta_1 \\ 6\beta_2 \end{pmatrix} \end{aligned}$$

**Theorem 0.6** (Vector-derivative of a product of matrices). If  $\mathbf{A}$  ( $\ell \times m$ ) and  $\mathbf{B}$  ( $m \times q$ ) are constant with respect to  $\tilde{\beta}$ , and  $\tilde{v} = \tilde{v}(\tilde{\beta})$  is a  $q \times 1$  vector that depends on the  $p \times 1$  vector  $\tilde{\beta}$ , then:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\mathbf{A}\mathbf{B}\tilde{v})}_{p \times \ell} = \underbrace{\left( \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right)}_{p \times q} \underbrace{\mathbf{B}^\top}_{q \times m} \underbrace{\mathbf{A}^\top}_{m \times \ell}$$

**i** Proof

*Proof.* Apply Theorem 0.5 with the constant  $\ell \times q$  matrix  $\mathbf{AB}$ , then use the [transpose of a product](#):

$$\begin{aligned}\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v}) &= \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) (\mathbf{AB})^\top \quad (\text{derivative of a matrix-vector product}) \\ &= \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{v} \right) \mathbf{B}^\top \mathbf{A}^\top \quad (\text{transpose of a product})\end{aligned}$$

□

**Example 0.8.** Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$  ( $1 \times 2$ ),  $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$  ( $2 \times 2$ ), and  $\tilde{v}(\tilde{\beta}) = \tilde{\beta}$  where  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ . Then  $\mathbf{AB}\tilde{v} = 2\beta_1$ , and:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\mathbf{AB}\tilde{v})}_{2 \times 1} = \underbrace{\left( \frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right)}_{2 \times 2} \underbrace{\mathbf{B}^\top \mathbf{A}^\top}_{2 \times 2 \times 1} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

**Corollary 0.1** (Derivative of a dot product, transpose-product form). *If  $\tilde{x}$  is constant with respect to  $\tilde{\beta}$ , then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta})}_{p \times 1} = \tilde{x}_{p \times 1}$$

*This vector derivative formula looks a lot like non-vector calculus, except that you have to transpose the coefficient: in scalar calculus  $\frac{\partial}{\partial x}(cx) = c$ , but here the coefficient  $\tilde{x}^\top$  (a row vector) becomes  $\tilde{x}$  (a column vector) in the result.*

**i** Proof

*Proof. Using Theorem 0.2:*

Since  $\tilde{x}^\top \tilde{\beta} = \tilde{x} \cdot \tilde{\beta}$  (see [dot product](#)), and  $\tilde{x}$  is constant with respect to  $\tilde{\beta}$ :

$$\frac{\partial}{\partial \tilde{\beta}}(\tilde{x}^\top \tilde{\beta}) = \frac{\partial}{\partial \tilde{\beta}}(\tilde{x} \cdot \tilde{\beta}) = \tilde{x}$$

by Theorem 0.2.

□

**i** Proof

*Proof.* Using **Theorem 0.5**:

Since  $\tilde{x}$  is constant with respect to  $\tilde{\beta}$ ,  $\mathbf{A} = \tilde{x}^\top$  is a constant  $1 \times p$  matrix. Applying **Theorem 0.5** with  $\tilde{v} = \tilde{\beta}$  (so  $\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} = \mathbf{I}$ ):

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} (\tilde{x}^\top \tilde{\beta}) &= \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right) (\tilde{x}^\top)^\top \\ &= \mathbf{I} \cdot \tilde{x} \\ &= \tilde{x} \end{aligned}$$

□

**Example 0.9** (Derivative of a transpose product). Let  $\tilde{x} = (3, 5)^\top$  and  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ . Then  $\tilde{x}^\top \tilde{\beta} = 3\beta_1 + 5\beta_2$ , and by **Corollary 0.1**:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} \left( \underbrace{\tilde{x}^\top}_{1 \times 2} \underbrace{\tilde{\beta}}_{2 \times 1} \right)}_{2 \times 1} = \underbrace{\tilde{x}}_{2 \times 1} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

**Theorem 0.7** (Derivative of a quadratic form). *For a quadratic form (see [quadratic form](#)), if  $\mathbf{S}$  is a symmetric  $p \times p$  matrix that is constant with respect to  $\tilde{\beta}$ , then:*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{p \times 1} = \underbrace{2\mathbf{S}\tilde{\beta}}_{p \times 1}$$

**i** Proof

*Proof.* Expanding entry-wise,  $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k$ . Differentiating component-wise with respect to  $\beta_i$  for  $i = 1, \dots, p$ :

$$\begin{aligned}
\left[ \frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) \right]_i &= \frac{\partial}{\partial \beta_i} \sum_{j=1}^p \sum_{k=1}^p s_{jk} \beta_j \beta_k \quad (\text{expand quadratic form}) \\
&= \sum_{k=1}^p s_{ik} \beta_k + \sum_{j=1}^p s_{ji} \beta_j \quad (\text{product rule for } \beta_i \beta_k) \\
&= [\mathbf{S} \tilde{\beta}]_i + [\mathbf{S}^\top \tilde{\beta}]_i \quad (\text{matrix-vector multiplication definition}) \\
&= [(\mathbf{S} + \mathbf{S}^\top) \tilde{\beta}]_i \quad (\text{linearity of matrix multiplication})
\end{aligned}$$

When  $\mathbf{S}$  is symmetric ( $\mathbf{S} = \mathbf{S}^\top$ ),  $\mathbf{S} + \mathbf{S}^\top = 2\mathbf{S}$ , so  $\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta}) = 2\mathbf{S} \tilde{\beta}$ .  $\square$

This operation is like taking the derivative of  $cx^2$  with respect to  $x$  in non-vector calculus.

**Example 0.10** (Derivative of a quadratic form). Let  $\mathbf{S} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$  ( $2 \times 2$ , symmetric and constant) and  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ . Then  $\tilde{\beta}^\top \mathbf{S} \tilde{\beta} = 3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2$ . By Theorem 0.7:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \mathbf{S} \tilde{\beta})}_{2 \times 1} = 2\mathbf{S} \tilde{\beta} = 2 \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Differentiating component-wise directly:

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \\ \frac{\partial}{\partial \beta_2} (3\beta_1^2 + 2\beta_1\beta_2 + 2\beta_2^2) \end{pmatrix} = \begin{pmatrix} 6\beta_1 + 2\beta_2 \\ 2\beta_1 + 4\beta_2 \end{pmatrix}$$

Both methods agree.

**Corollary 0.2** (Derivative of a simple quadratic form).

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}} (\tilde{\beta}^\top \tilde{\beta})}_{p \times 1} = \underbrace{2\tilde{\beta}}_{p \times 1}$$

### **i** Proof

*Proof.* Applying Theorem 0.7 with  $\mathbf{S} = \mathbf{I}_{p \times p}$  (which is symmetric and constant with respect to  $\tilde{\beta}$ ):

$$\begin{aligned}
\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta}) &= \frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \mathbf{I}_{p \times p} \tilde{\beta}) \quad (\text{rewrite with identity matrix}) \\
&= 2\mathbf{I}_{p \times p} \tilde{\beta} \quad (\text{derivative of a quadratic form, with } \mathbf{S} = \mathbf{I}_{p \times p}) \\
&= 2\tilde{\beta} \quad (\text{identity matrix property})
\end{aligned}$$

□

This vector derivative is like taking the derivative of  $x^2$ .

**Example 0.11** (Derivative of a sum of squares). Let  $\tilde{\beta} = (\beta_1, \beta_2)^\top$ , so  $\tilde{\beta}^\top \tilde{\beta} = \beta_1^2 + \beta_2^2$ . By Corollary 0.2:

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\beta}^\top \tilde{\beta})}_{2 \times 1} = 2\tilde{\beta} = \begin{pmatrix} 2\beta_1 \\ 2\beta_2 \end{pmatrix}$$

Direct partial differentiation yields the same column vector.

**Theorem 0.8** (Vector chain rule). *Let  $\tilde{x}$  be a  $p \times 1$  vector, let  $\tilde{y} = \tilde{g}(\tilde{x})$  be a  $q \times 1$  vector-valued function of  $\tilde{x}$ , and let  $z = f(\tilde{y})$  be a scalar-valued function of  $\tilde{y}$ , where  $\tilde{g}$  and  $f$  have continuous partial derivatives. Then  $z = f(\tilde{g}(\tilde{x}))$ , as a function of  $\tilde{x}$ , satisfies*

$$\underbrace{\frac{\partial z}{\partial \tilde{x}}}_{p \times 1} = \underbrace{\frac{\partial \tilde{y}}{\partial \tilde{x}}}_{p \times q} \underbrace{\frac{\partial z}{\partial \tilde{y}}}_{q \times 1}$$

where  $\frac{\partial \tilde{y}}{\partial \tilde{x}}$  is the derivative of Definition 0.3 and  $\frac{\partial z}{\partial \tilde{x}}$  and  $\frac{\partial z}{\partial \tilde{y}}$  are vector derivatives (Definition 0.1).

See <https://quickfem.com/finite-element-analysis/>, specifically [https://quickfem.com/wp-content/uploads/IFEM.AppF\\_.pdf](https://quickfem.com/wp-content/uploads/IFEM.AppF_.pdf)

See also [https://en.wikipedia.org/wiki/Gradient#Relationship\\_with\\_Fr%C3%A9chet\\_derivative](https://en.wikipedia.org/wiki/Gradient#Relationship_with_Fr%C3%A9chet_derivative)

This chain rule is like the univariate [chain rule](#), but the order matters now. The version presented here is for the [gradient](#) (column vector); the [total derivative](#) (row vector) would be the [transpose of the gradient](#).

**Corollary 0.3** (Vector chain rule for quadratic forms). *If  $\tilde{\varepsilon} = \tilde{\varepsilon}(\tilde{\beta})$  is an  $n \times 1$  vector-valued function of the  $p \times 1$  vector  $\tilde{\beta}$ , with continuous partial derivatives, then*

$$\underbrace{\frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon}(\tilde{\beta}) \cdot \tilde{\varepsilon}(\tilde{\beta}))}_{p \times 1} = \underbrace{\left( \frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon}(\tilde{\beta}) \right)}_{p \times n} \underbrace{(2\tilde{\varepsilon}(\tilde{\beta}))}_{n \times 1}$$

### **i** Proof

*Proof.* Apply Theorem 0.8 with  $\tilde{x} = \tilde{\beta}$ ,  $\tilde{y} = \tilde{\varepsilon}$ ,  $q = n$ , and  $z = f(\tilde{\varepsilon}) = \tilde{\varepsilon} \cdot \tilde{\varepsilon} = \tilde{\varepsilon}^\top \tilde{\varepsilon}$ :

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) \frac{\partial}{\partial \tilde{\varepsilon}}(\tilde{\varepsilon}^\top \tilde{\varepsilon}) \quad (\text{vector chain rule}) \\ &= \left( \frac{\partial}{\partial \tilde{\beta}} \tilde{\varepsilon} \right) (2\tilde{\varepsilon}) \quad (\text{derivative of a simple quadratic form, in } \tilde{\varepsilon}) \end{aligned}$$

The second step is Corollary 0.2, applied to the  $n \times 1$  vector  $\tilde{\varepsilon}$  in place of  $\tilde{\beta}$ .  $\square$

**Example 0.12** (Derivative of the residual sum of squares). Let  $\tilde{y}$  ( $n \times 1$ ) and  $\mathbf{X}$  ( $n \times p$ ) be constant with respect to  $\tilde{\beta}$ , and let  $\tilde{\varepsilon}(\tilde{\beta}) = \tilde{y} - \mathbf{X}\tilde{\beta}$  be the vector of residuals. By Theorem 0.4,  $\frac{\partial}{\partial \tilde{\beta}}(\mathbf{X}\tilde{\beta}) = \mathbf{X}^\top$ , and  $\frac{\partial}{\partial \tilde{\beta}}\tilde{y} = \mathbf{0}_{p \times n}$  because  $\tilde{y}$  is constant, so  $\frac{\partial}{\partial \tilde{\beta}}\tilde{\varepsilon} = -\mathbf{X}^\top$  ( $p \times n$ ). By Corollary 0.3:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}}(\tilde{\varepsilon} \cdot \tilde{\varepsilon}) &= (-\mathbf{X}^\top)(2\tilde{\varepsilon}) \quad (\text{vector chain rule for quadratic forms}) \\ &= -2\mathbf{X}^\top(\tilde{y} - \mathbf{X}\tilde{\beta}) \quad (\text{substitute } \tilde{\varepsilon} = \tilde{y} - \mathbf{X}\tilde{\beta}) \end{aligned}$$

Setting this  $p \times 1$  vector to  $\tilde{0}$  gives the normal equations  $\mathbf{X}^\top \mathbf{X}\tilde{\beta} = \mathbf{X}^\top \tilde{y}$  of least squares.

**Definition 0.5** (Matrix derivative). For a scalar-valued function  $f(\mathbf{X})$  of an  $m \times n$  matrix  $\mathbf{X}$ , the **matrix derivative** is the  $m \times n$  matrix whose  $(i, j)$  entry is the partial derivative of  $f$  with respect to the  $(i, j)$  entry of  $\mathbf{X}$ :

$$\left[ \frac{\partial}{\partial \mathbf{X}} f \right]_{ij} = \frac{\partial}{\partial X_{ij}} f$$

**Example 0.13** (The matrix derivative of a trace). Let  $\mathbf{X}$  be a  $2 \times 2$  matrix and  $f(\mathbf{X}) = \text{tr}(\mathbf{X}) = X_{11} + X_{22}$  (see [trace](#)). Then  $\frac{\partial}{\partial X_{ij}} f = 1$  if  $i = j$  and 0 otherwise, so:

$$\frac{\partial}{\partial \mathbf{X}} f = \mathbf{I}_2$$

**Theorem 0.9** (Matrix derivative of the trace of a matrix product). *If  $\mathbf{A}$  ( $r \times m$ ) and  $\mathbf{B}$  ( $n \times r$ ) are constant with respect to the  $m \times n$  matrix  $\mathbf{X}$ , then:*

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{m \times n} = \underbrace{\mathbf{A}^\top}_{m \times r} \underbrace{\mathbf{B}^\top}_{r \times n}$$

*The trace makes  $\text{tr}(\mathbf{AXB})$  a scalar, so its matrix derivative is again an  $m \times n$  matrix. The derivative of the matrix product  $\mathbf{AXB}$  itself (without the trace) is a fourth-order tensor, which is why this result is stated for the scalar  $\text{tr}(\mathbf{AXB})$ .*

### **i** Proof

*Proof.* Write  $A_{kl}$ ,  $X_{kl}$ , and  $B_{kl}$  for the entries of  $\mathbf{A}$ ,  $\mathbf{X}$ , and  $\mathbf{B}$ . For entry  $(i, j)$ :

$$\begin{aligned} \left[ \frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB}) \right]_{ij} &= \frac{\partial}{\partial X_{ij}} \sum_{a=1}^r \sum_{b=1}^m \sum_{c=1}^n A_{ab} X_{bc} B_{ca} \quad (\text{trace of the } r \times r \text{ product } \mathbf{AXB}) \\ &= \sum_{a=1}^r A_{ai} B_{ja} \quad (\text{only the terms with } b = i, c = j \text{ depend on } X_{ij}) \\ &= \sum_{a=1}^r [\mathbf{A}^\top]_{ia} [\mathbf{B}^\top]_{aj} \quad (\text{definition of the transpose}) \\ &= [\mathbf{A}^\top \mathbf{B}^\top]_{ij} \quad (\text{definition of matrix multiplication}) \end{aligned}$$

□

**Example 0.14** (Differentiating a weighted trace). Let  $\mathbf{A} = \mathbf{I}_2$  ( $2 \times 2$ ) and  $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$  ( $2 \times 2$ ). Then  $\text{tr}(\mathbf{AXB}) = 2X_{11} + 3X_{22}$ , and:

$$\underbrace{\frac{\partial}{\partial \mathbf{X}} \text{tr}(\mathbf{AXB})}_{2 \times 2} = \underbrace{\mathbf{A}^\top}_{2 \times 2} \underbrace{\mathbf{B}^\top}_{2 \times 2} = \mathbf{I}_2 \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

**Exercise 0.2** (A gradient that does not mention its variable). Fix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix}$$

let  $W \in \mathbb{R}^{2 \times 3}$  vary, and define

$$g(W) = \sum_{i=1}^2 \sum_{j=1}^3 W_{ij} A_{ij}$$

which is the entry-by-entry version of the inner product for matrices. Compute  $\nabla_W g(W)$ , and say what is unusual about the answer.

### Solution

*Solution 0.2.* Written out over the six positions,

$$g(W) = 2W_{11} - W_{12} + 0W_{13} + 4W_{21} + 3W_{22} - 2W_{23}$$

Each  $W_{ij}$  appears in exactly one term, multiplied by  $A_{ij}$  and by nothing else, so differentiating with respect to it leaves  $A_{ij}$  behind:

$$\frac{\partial g}{\partial W_{ij}} = A_{ij}$$

Collecting those partial derivatives into a matrix of the same shape as  $W$ , as Table 1 requires,

$$\nabla_W g(W) = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & -2 \end{bmatrix} = A$$

What is unusual is that the gradient came out **constant**: it does not mention  $W$  at all. The constant answer is not a quirk of this particular  $A$ .  $g$  is a linear function of  $W$ , and the gradient of a linear function is constant everywhere, for the same reason the derivative of  $f(w) = cw$  is  $c$  no matter where it is evaluated.

A constant gradient is the easy case, and it is not the case we usually face. Most objectives are curved — the squared error of a linear model is the standard example — so their gradient changes from point to point and the downhill direction has to be worked out afresh at every step.

### Additional resources

See also the [Linear Algebra and Vector Calculus](#) references.

- [Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" \(2023 Fall\)](#)

## References

- Fieller, Nick. 2016. *Basics of Matrix Algebra for Statistics with R*. Chapman; Hall/CRC. <https://doi.org/10.1201/9781315370200>.
- Hutchinson, Brian. n.d. *DATA 471/571 (Machine Learning) and CSCI 481/581 (Deep Learning) Video Lectures*. Western Washington University. Accessed September 28, 2026. [https://facultyweb.cs.wvu.edu/~hutchib2/video\\_lectures/data371/](https://facultyweb.cs.wvu.edu/~hutchib2/video_lectures/data371/).